

From Psychology Research Articles to Structured Knowledge: A Reproducible LLM Pipeline for Knowledge Graph Construction and Analysis

Lay Reader Background

Psychology research produces vast amounts of knowledge about human behavior, emotions, and cognition. However, most of this knowledge is buried in text-heavy research papers, making it difficult to search, compare, or reuse across studies. For example, the same concept (like “self-esteem”) may be studied using different measurement tools, in different populations, and under different conditions—yet these relationships are rarely captured in a structured way.

Recent advances in artificial intelligence, particularly large language models (LLMs), make it possible to automatically extract such information from text. By converting unstructured research articles into structured data, we can build knowledge graphs that organize psychological knowledge in a machine-readable and reusable format.

Scientific Background

Psychological science faces a well-known challenge of **conceptual fragmentation**, where constructs are inconsistently defined and operationalized across studies. This leads to issues such as **jingle-jangle fallacies**, limiting cumulative knowledge building.

At the same time, research workflows remain largely **narrative-driven**, relying on manual literature reviews and subjective interpretation. While LLMs have shown strong capabilities in extracting structured information from text, existing approaches typically focus on minimal schemas (e.g., construct–measure pairs) and lack:

- contextual information (population, setting)
- methodological details (statistics, reliability)
- normalization and alignment of constructs

Recent work demonstrates that schema-guided LLM pipelines, combined with validation frameworks (e.g., Pydantic), enable **robust structured extraction** from scientific text (refer to the two blogposts in the references sections). However, there is a need to extend these pipelines toward richer schemas, systematic evaluation, and downstream analysis such as embedding-based exploration and construct normalization.

Aim

The aim of this thesis is to design and evaluate a **schema-driven, reproducible pipeline** for extracting structured psychological knowledge from research articles and representing it in a knowledge graph.

Specifically, the thesis aims to:

- develop an LLM-based extraction pipeline aligned with an extended schema
- populate a structured knowledge graph from full-text psychology articles
- explore embedding-based representations for visualization and analysis
- investigate methods for construct normalization and disambiguation using textual evidence

Knowledge Extraction Schema

```
class PsychTriple(BaseModel):
```

```
    construct: str          # raw construct mention from text
```

```
    normalized_construct: str | None # canonical/standardized construct name
```

```
    measured_by: str        # instrument/task used to measure the construct
```

```
    population: str | None  # study sample (who was measured)
```

```
    setting: str | None     # experimental or data collection context
```

```
    statistic: str | None   # reported quantitative result (e.g.,  $r=0.45$ ,  $\beta=0.32$ )
```

```
    reliability: str | None # measurement reliability (e.g., Cronbach's alpha)
```

```
    justification: str      # text snippet supporting the extraction
```

Property Definitions

- **construct**: The psychological concept as mentioned in the paper (may be noisy or varied).
- **normalized_construct**: A standardized version of the construct (for aligning synonyms).
- **measured_by**: The measurement instrument, task, or method used.
- **population**: The group of participants studied (e.g., adults, patients).
- **setting**: The context in which data was collected (e.g., lab, survey, clinical).
- **statistic**: A reported quantitative result linking construct and measurement.
- **reliability**: Indicator of measurement quality (e.g., internal consistency).
- **justification**: Evidence text from the paper supporting the extracted information.

Example Instantiations

Example 1

```
{
  "construct": "self-esteem",
  "normalized_construct": "self-esteem",
  "measured_by": "Rosenberg Self-Esteem Scale",
  "population": "undergraduate students",
  "setting": "self-report survey",
  "statistic": "r = 0.52",
  "reliability": "Cronbach's alpha = 0.88",
  "justification": "Self-esteem was assessed using the Rosenberg Self-Esteem Scale ( $\alpha = .88$ )."
```

Example 2

```
{
  "construct": "working memory capacity",
  "normalized_construct": "working memory",
  "measured_by": "n-back task",
  "population": "healthy adults",
  "setting": "laboratory experiment",
  "statistic": null,
  "reliability": null,
  "justification": "Participants performed a 2-back task to assess working memory capacity."
```

Example 3

```
{
  "construct": "depressive symptoms",
  "normalized_construct": "depression",
  "measured_by": "Beck Depression Inventory-II",
  "population": "patients with major depressive disorder",
  "setting": "clinical assessment",
  "statistic": " $\beta = 0.34$ ",
  "reliability": " $\alpha = 0.91$ ",
  "justification": "Depressive symptoms were measured using the BDI-II ( $\alpha = .91$ ), predicting outcomes significantly ( $\beta = .34$ )."
```

Research Questions

RQ1 — Knowledge extraction pipeline (KG construction)

How can a reproducible LLM-based pipeline be designed to extract schema-complete psychological knowledge (constructs, measures, context, and evidence) from full-text articles and populate a consistent knowledge graph?

RQ2 — Embedding-based visualization

How can embedding-based representations of constructs and measurement instruments—individually and jointly—be used to visualize and explore the structure of psychological knowledge graphs?

RQ3 — Description-based normalization & disambiguation

How can construct normalization be improved by leveraging textual descriptions (justifications) to disambiguate and align semantically similar or overlapping psychological constructs?

Key implications for behavioral sciences

- **Clearer constructs**
Makes construct–measure links explicit → reduces conceptual confusion.
- **Better evidence synthesis**
Enables scalable, machine-readable comparisons and meta-analysis.
- **New analysis possibilities**
Embeddings + KG reveal structure and relationships across studies.
- **More reproducibility**
Standardized extraction reduces subjective interpretation.
- **FAIR knowledge**
Turns papers into reusable, machine-actionable data.
- **Caveat**
Needs validation to avoid propagating errors or oversimplification.

👉 Bottom line:

This shifts behavioral science from *narrative-driven knowledge* → **structured, computable, and comparable science**.



Conclusion

This work contributes to bridging the gap between **unstructured scientific literature and structured, machine-actionable knowledge**. By combining schema-guided extraction, validation, and embedding-based analysis, the thesis demonstrates how modern AI methods can support more transparent, scalable, and systematic research practices in psychology.



Expected Outcome

The outcome of this thesis is:

- a **fully reproducible pipeline** for structured knowledge extraction
- a **curated dataset / knowledge graph** of psychological constructs and measures
- an **evaluation of extraction, embedding, and normalization methods**
- and a **tutorial-style methodological paper** suitable for submission to *Behavior Research Methods* (<https://link.springer.com/journal/13428>), demonstrating how behavioral scientists can build and use such pipelines in practice

References

<https://medium.com/@jenlindadsouza/psychkg-how-to-build-a-minimal-knowledge-graph-for-psychology-fac0c76800ac>
<https://medium.com/@jenlindadsouza/how-i-get-llms-on-hugging-face-to-speak-structured-data-1fb34bf15792>
<https://github.com/sciknoworg/psychKG-pilot/tree/main>